

Michael Lin

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Education

Carnegie Mellon University

M.S. in Computer Vision (Robotics Institute)

Pittsburgh, PA

August 2025 – December 2026

- **Relevant coursework:** Machine Learning, Advanced Computer Vision, Learning for 3D Vision, SLAM, Robot Learning, Visual Learning and Recognition

Boston College

B.S. in Computer Science, B.A. in Mathematics

Chestnut Hill, MA

August 2021 – May 2025

Experience

Computer Vision Student Researcher (Near Earth Autonomy)

January 2026 – present

- Investigated and implemented improvement modules on YOLOv8-seg, leading to a 4% improvement in mAP for monocular runway segmentation from simulated fixed-wing aircraft over baseline YOLO
- Developed a joint estimator for aircraft yaw and cross-track distance by optimizing segmentation masks and fusing predictions using extended Kalman filtering, achieving a maximum of 0.4 degrees of yaw error across test suite of 10+ flights
- Built a per-frame geometry-based estimator of cross-track distance using bird's-eye view transforms and Kalman filtering, achieving an average of 1.22 meters of error across test suite of 120+ landing sequences

Machine Learning Validation Intern (General Motors / Cruise)

May 2026 – August 2026

- Developed AI agent skills to search national-scale road map with natural language descriptions using Model Workspace Protocol (MWP) design and Google BigQuery. Achieved over 70% precision during testing on a suite of over twenty varied descriptions and reduced development time of specialized geometry-based map queries from over one week to approximately one hour
- Prototyped AI-based pipeline for categorization of type and severity of inclement weather using a lightweight CNN, achieving up to 90% precision and recall on a curated testing set of over 100 drives

Undergraduate Research Fellow (Boston College)

January 2022 – August 2025

- Developed UNet-based machine learning pipelines for cell segmentation in electron microscopy imagery over multiple projects. Fine-tuned and scaled models during deployment to generate segmentation maps on terabytes of data. Advised by Professor Donglai Wei and collaborated with labs at USC, Columbia University, and UMichigan to publish multiple papers in high-profile biology journals (Nature, Cell)

Projects

Visual Odometry and SLAM in Game Environments

January 2026 – May 2026

- Benchmarked classical and learning-based visual odometry and SLAM methods (ORB-SLAM3, MAST3R-SLAM, DPVO, TartanVO) on 572 trajectories across 31 photorealistic video game scenes from the OmniWorld dataset. Built a unified pipeline for data conversion and trajectory export, and used Sim(3)-aligned ATE and RPE to evaluate global consistency and local drift
- Investigated DPVO's patch selection under fast camera motion and applied region masking to remove the sky and player model from tracking, which reduced translational RPE by up to 19% on a challenging scene

Open-Vocabulary Scene Understanding for 3D Gaussian Splatting

September 2025 – December 2025

- Developed a 3D Gaussian splatting tool based on OpenGaussian and StreetGaussian for zero-shot semantic understanding. Used 2D segmentation masks from Segment Anything to facilitate learning of a discrete codebook of open vocabulary language embeddings and utilized cosine distance as a guiding metric for answering natural language queries

Technical Skills

Languages: English (native proficiency), Mandarin Chinese (conversational proficiency)

Programming Languages: Python, C, C++, Java, R, SQL

Tools: Pytorch, OpenCV, Numpy, PIL, Scikit, Matplotlib, Scipy, Git, Slurm, Github, Linux, Jupyter, Optuna